Problem Solving through Programming (CSE103)
Multiple Choice Questions
Unit-3

Q.No		
1`	What is the index of the first element in an array?	
	a) 0	b) 1
	c) First	d) Start
2	In an array declaration int arr[5] = {1, 2, 3};, what will be the value of arr[3]?	
	a) 1	b) 2
	c) 3	**d) 0**
3	Which statement about arrays in C is false?	
	a) Arrays can store multiple values of different data types.	b) The size of an array must be specified at the time of declaration.
	c) The elements of an array are stored in contiguous memory locations.	d) The array index starts from 0.
4	How do you access the last element of an array arr with n elements in C	
	a) arr[n];	**b) arr[n-1];**
	c) arr[n+1];	d) arr[last];
5	Assuming int is of 4bytes, what is the size of int arr[15];?	
	a) 15	b) 19
	c) 11	**d) 60**
6	To access the third element of an array named ‘arr’, you would use:	
	a) arr[2]	B) arr[3]
	c) arr(2)	d) arr(3)
7	Which of the following correctly initializes an array of size 5 with all values set to 0?	
	a) int arr[5] = {0};	b) int arr[5];
	c) int arr[5] = {};	d) int arr[] = {0,0,0,0,0};
8	How would you declare a two-dimensional integer array of 3 rows and 4 columns?	
	a) int arr[3, 4];	**b) int arr[3][4];**
	c) int[3][4] arr;	d) int arr[ ][ ] = {3, 4};

9	What will be the output of the following C code?
	
	#include <stdio.h>
	int main()
	{
	int ary[2][3];
	ary[][] = {{1, 2, 3}, {4, 5, 6}};
	printf("%d\n", ary[1][0]);
	}	

	a) Compile time error	b) 4
	c) 1	d) 2
10	Given a 2D array int a[2][3] = {{1, 2, 3}, {4, 5, 6}};, what does a[1][2] return?	
	a) 3	b) 4
	c) 5	**d) 6**
11	Which operation on a two-dimensional array requires nested loops in C?	
	a) Printing all elements	b) Initializing all elements to zero
	c) Summing all elements	**d) All of the above**
12	What is the correct declaration of a function that accepts a two-dimensional array with 3	

	columns in C?	
	a) void func(int arr[][])	b) void func(int arr[3][])
	c) void func(int arr[][3])	d) void func(int arr[3][3])
13	How many passes will bubble sort make to sort the array {5, 1, 4, 2, 8}?	
	a) 3	**b) 4**
	c) 5	d) 2
14	In linear search, how are elements checked?	
	a) By checking from the middle	**b) By checking from first to last one by one**
	c) By checking from last to first only	d) Randomly checking elements
15	Binary search finds the number by:	
	a) Checking all elements one by one	**b) Dividing the list into halves each time**
	c) Starting from the last element	d) Jumping to random elements
16	Bubble sort works by:	
	a) Swapping the biggest element to the end in each pass	b) Picking the smallest and putting it first
	c) Randomly swapping elements	d) Sorting only half of the list
17	After the first pass in bubble sort, the largest element is:	
	a) At the start of the array	b) In the middle
	c) At the end of the array	d) Deleted from the array
18	In selection sort, what is selected in each pass?	
	a) The largest element	**b) The smallest element**
	c) Any random element	d) Middle element

19	int arr[] = {4, 7, 9};
	int key = 7;
	for(int i=0; i<3; i++) {
	if(arr[i] == key) {
	printf("%d", i);
	break;
	}
	}	

	a) 0	**b) 1**
	c) 2	d) Nothing
20	If the number we are looking for is at the first position in the array, linear search will take:	
	a) Many steps	**b) Only one step**
	c) No steps	d) Exactly half the steps
21`	A string constant in C is terminated by	
	a) '\0'	b) "\0"
	c)' '	d)" "
22	What will strcmp() function do?	
	a) compares the first n characters of the object	b) undefined function
	c) copies the string	**d) compares the string**
23	Which of the following function is appropriate for reading a multi-word string?	
	a) printf()	b) scanf()
	c) gets()	d) puts()
24	The ---------- function appends not more than n characters.	
	a) strcat()	**b) strncat()**

	c) memcat()	d) strcpy()
25	When strcpy(str1,str2) is executed,	
	a) str1 is stored in str2	**b) str2 is stored in str1**
	c)copied string is stored in virtual memory	d) None of the above
26	A C string elements are always stored in?	
	a) Sequential memory locations	b) Random memory locations
	c)Does not have memory location	d) Alternate memory locations
27	If the two strings are identical, then strcmp() function returns	
	a)-1	b)1
	c)0	d)NULL
28	How do you find the length of a string in C?	
	a) Using the length() function	**b) Using the strlen() function**
	c) Using the sizeof operator	d) By manually counting the characters
29	What happens if a string array is initialized with fewer characters than its size in C?	
	a) The program crashes	b) It causes an error
	c) The remaining elements are left uninitialized	**d) The remaining elements are filled with null characters**
30	Which header file must be included to use string functions like strcpy() and strcat() in C?	
	a) #include <string.h>	b) #include <stdio.h>
	c) #include <stdlib.h>	d) #include <strings.h>